AEROSPACE ENGINEERING · CAL POLY POMONA · EXPECTED MAY 2029

Edwin Calles Molina

Aerospace Engineering student at Cal Poly Pomona. I design rockets, build them, fly them, and work from the data that comes back.

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4,135 ft
Level 2 certification flight apogee

Mach 1.2
peak speed on my second rocket

4
documented flight tests on 2 vehicles

3.97
GPA · President's Honor List

I'm a second-year Aerospace Engineering student at Cal Poly Pomona with a 3.97 GPA, NAR Level 2 High Power Rocketry certification, and four documented flight tests behind me. I designed these rockets in SolidWorks, 3D-printed and assembled the hardware, assembled and operated the data acquisition and GPS, and flew them.

On my most recent flight I reconstructed a supersonic structural failure from the recovered flight record, caught a corrupted altimeter reading in the process, validated the real trajectory against backup inertial data, and specified the redesign that followed.

FLIGHT LOG

FLIGHT	DATE	VEHICLE	APOGEE	MAX VELOCITY	OUTCOME
L2 — first attempt	Jul 18, 2026	Rocket 1	4,224 ft	659 ft/s	Failure Level 2 attempt failed. Descent rate went from 44 to 46 ft/s at main deployment instead of slowing. Fins damaged. Recovery train redesigned.
L2 — certification flight	Aug 8, 2026	Rocket 1 (rebuilt)	4,135 ft	658 ft/s	Nominal NAR Level 2 certification earned on a J250W motor (~704 N·s). 6.8° tilt at burnout.
Subsonic shakedown	Aug 8, 2026	Rocket 2	2,833 ft	481 ft/s	Nominal Recovery nominal, both events on time. Vehicle cleared for the supersonic attempt.
Supersonic attempt	Aug 15, 2026	Rocket 2	Breakup at ≈ 1,800 ft	1,367 ft/s (Mach 1.2)	Failure Structural failure at t = 2.20 s. Fin failure identified as the initiating event. Redesign specified.

IN THIS PORTFOLIO

Dual-deploy high power rocket — NAR Level 2 certification — INDEPENDENT PROJECT · ROCKET 1 · JUNE – AUGUST 2026

Supersonic flight test and failure reconstruction — INDEPENDENT PROJECT · ROCKET 2 · AUGUST 2026

Bronco Avionics — Co-lead, Launch Operations — BRONCO LAUNCH · BRONCO AVIONICS · CAL POLY POMONA · JOINED MARCH 2026 · CO-LEAD SINCE MAY 2026

Small Rocket Competition — sub-team lead — UMBRA, NOW BRONCO LAUNCH · CAL POLY POMONA · SEPTEMBER – DECEMBER 2025

SR-30 turbojet performance test — CAL POLY POMONA · FALL 2025

Flight log · Now · Skills · About — REFERENCE

U.S. citizen · eligible for a U.S. government security clearance. Available June 2027 · Summer 2027 engineering internship · willing to relocate · onsite five days a week. Flight figures are from the Featherweight Blue Raven 2.0.

INDEPENDENT PROJECT · ROCKET 1 · JUNE – AUGUST 2026

Dual-deploy high power rocket — NAR Level 2 certification

3.1 in OD / 3.0 in ID Blue Tube airframe · 54 mm motor mount · AeroTech J250W, ~704 N·s · 3D-printed PETG ogive nosecone · birch plywood clipped-delta fins, 4.30 in root chord · dual-deploy recovery, 48 in main · Featherweight Blue Raven 2.0 · Eggfinder GPS · two flights

ROLE	Designer, builder, and test operator. I designed the vehicle and the electronics bay, wired and soldered the recovery electronics, and flew it.
STATUS	Certified · NAR Level 2 High Power Rocketry, August 2026
TOOLS	SolidWorks · OpenRocket · FDM 3D printing · Soldering and wiring harness assembly · Dual-deploy recovery design · Blue Raven 2.0 altimeter setup · Ejection charge sizing · Flight data analysis

4,135 ft
apogee on the Level 2 certification flight

6.8°
tilt from vertical at burnout

1,498 ft
main deployment · programmed for 1,500 ft

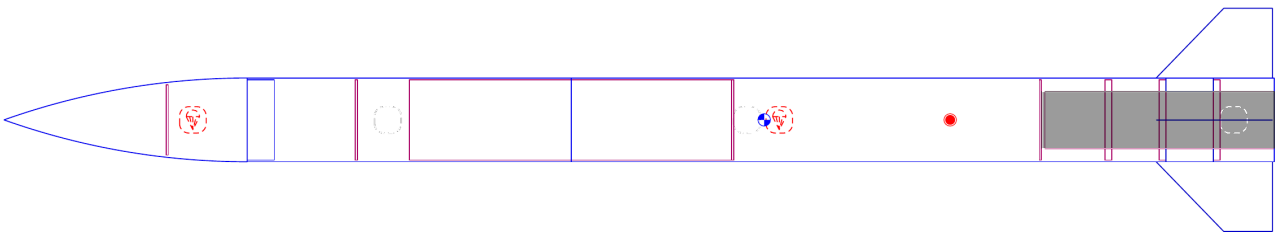
19 ft/s
descent rate under the main

59.6 G
touchdown load the printed bay survived

My first Level 2 attempt failed right in front of me: the drogue came out, the main never did. I found the cause, restaged it, and flew again. The second flight went to 4,135 feet and earned my Level 2.

OBJECTIVE

Gain experience designing, manufacturing, and constructing a rocket capable of withstanding a launch on a motor between 640 and 5,120 newton-seconds. Obtain the Level 2 High Power Rocketry certification. Obtain and analyze flight data to improve the design and construction of the next vehicle.



The Level 2 vehicle in OpenRocket: 9 in printed PETG nosecone, 12 in upper body tube over the coupler and avionics bay, main chute ahead of it, and the 26 in lower body tube on a 54 mm motor mount. The center of gravity sits 2.2 calibers ahead of the center of pressure.

CHALLENGES FACED

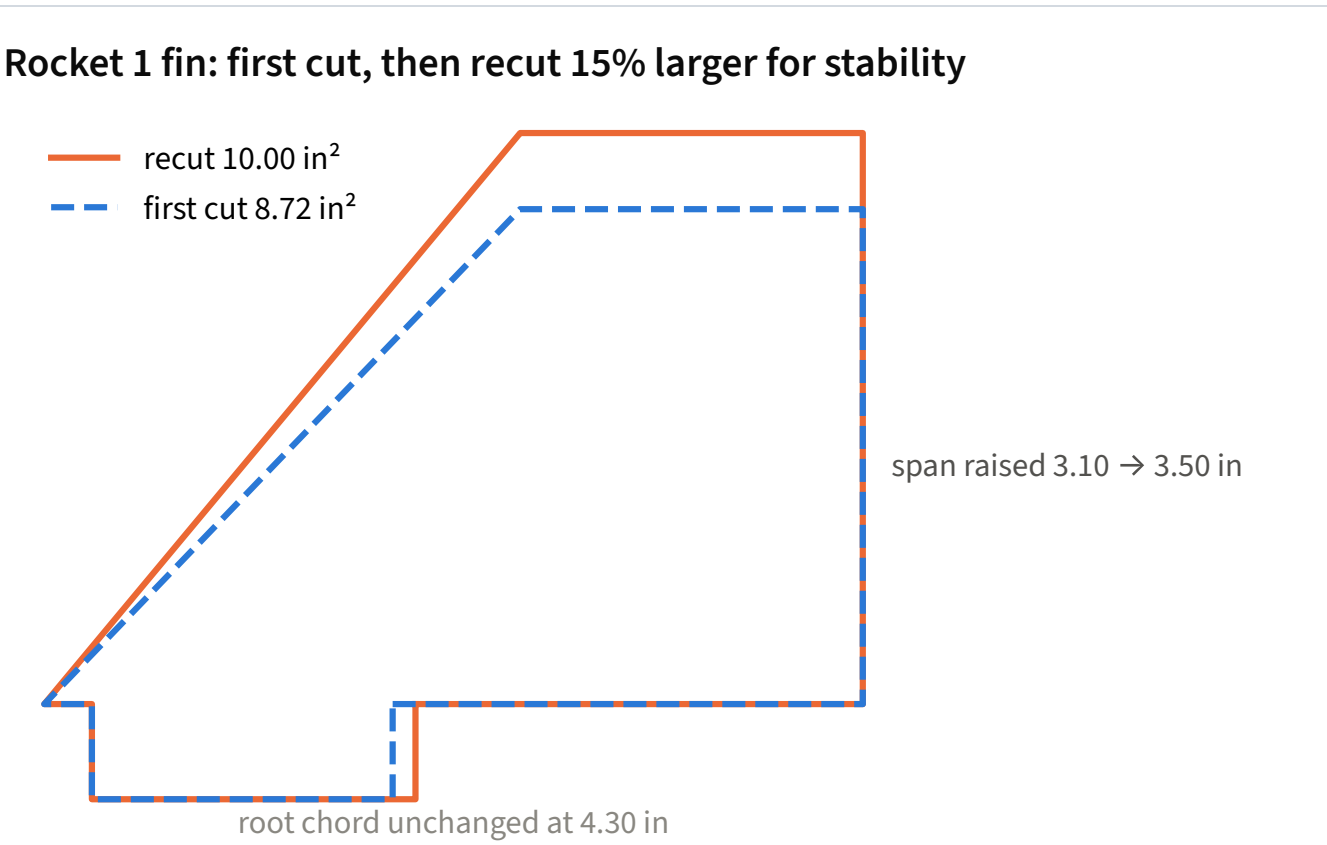
There were three primary challenges during the designing and constructing phase of an L2-capable vehicle.

1. Using a motor ~4x as powerful as my previous strongest meant using stronger materials and a more refined construction.
2. Brand new electronics I had little experience with meant hours researching, tinkering, and soldering.

3. And finally, stepping the recovery system up from a single-deploy motor-actuated deployment to a dual-deploy electronic recovery meant I had to research and correctly place the new recovery system to prevent damage during recovery.

SIZING THE FINS

I cut a first set of fins, then went back to the simulation and found the stability margin sitting lower than I wanted, which meant the fins were simply too small for the airframe. I kept the root chord at 4.30 inches and raised the span from 3.10 to 3.50 inches, taking the fin area from 8.72 to 10.00 square inches. That is a 15 percent increase in area, and it brought the static margin up near the 2 calibers I was aiming for.



The Rocket 1 fin as first cut and as recut. Same 4.30 in root chord, span raised from 3.10 to 3.50 in, which is 15 percent more area and the stability margin I needed.

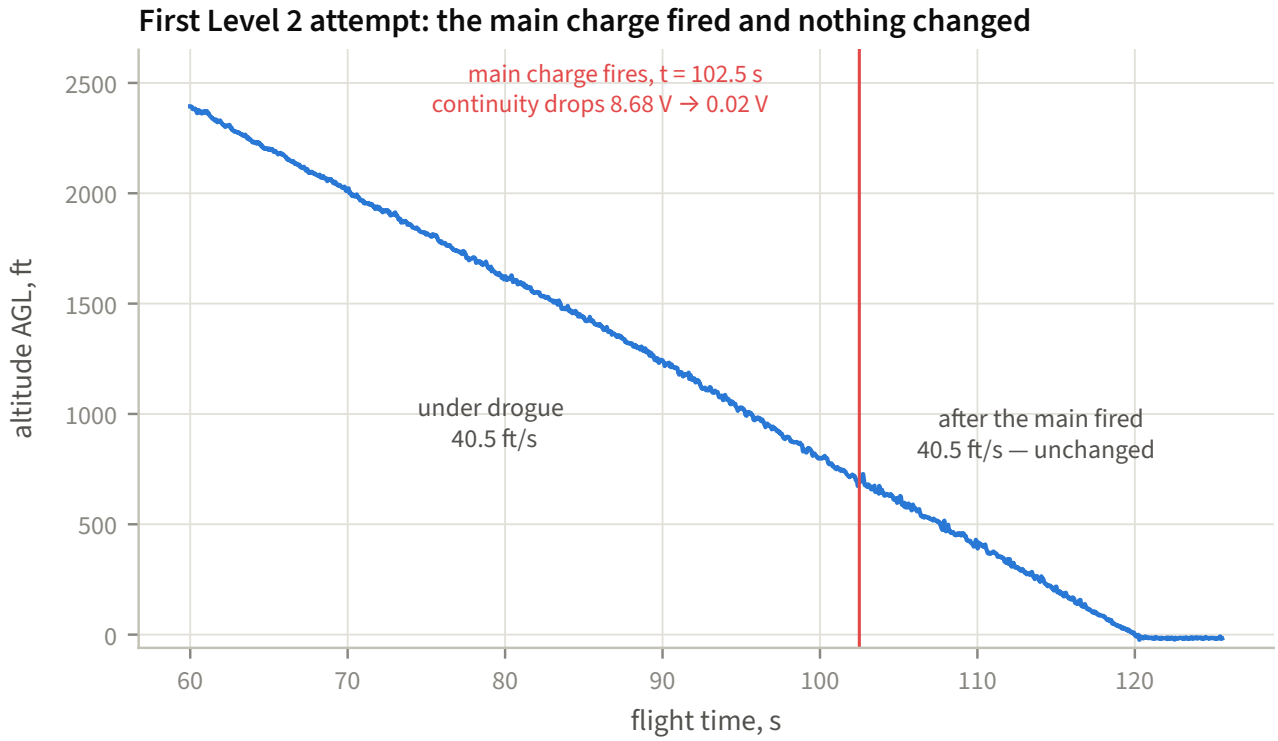
L2 — FIRST ATTEMPT AT CERTIFICATION

Over time, I designed, built, and flew the first iteration of the L2-capable rocket. The result? The main chute never opened. An incorrect separation during recovery prevented it from deploying at its programmed altitude. Luckily, the drogue chute worked — enough to save the airframe, but not the fins. I was frustrated. Half my summer was sitting around me in pieces.

The rocket flew on 18 July 2026 at the Lucerne Valley lakebed with the Rocketry Organization of California, using the same AeroTech J250W I would later fly for the Level 2 certification, and it reached 4,224 feet. The flight record says the rest of it plainly. The main charge fired where it was supposed to, at 102.5 seconds and 694 feet, and the continuity voltage on that channel drops from 8.68 volts to 0.02 volts inside a single 20 millisecond sample, which is the e-match burning through. What did not change was the descent rate. The rocket was coming down at 40.5 feet per second under the drogue

before the charge fired, and it was still coming down at 40.5 feet per second afterward, meaning the canopy never inflated and the vehicle rode the drogue the whole way to the ground.

It landed hard enough to break the fins. One came off the airframe entirely, a second was damaged, a third cracked at the root, and the fourth was untouched.

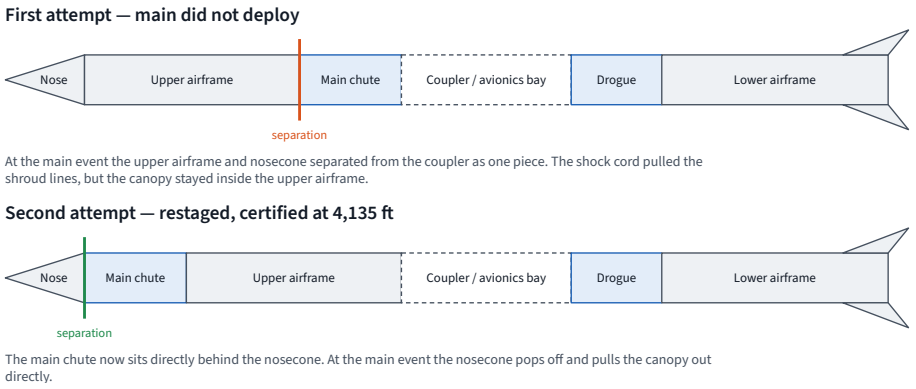


The first Level 2 attempt, from the barometric channel. The main charge fired at 102.5 seconds and the descent rate on either side of it is the same number, 40.5 ft/s, meaning the canopy never inflated.

FINDING THE CAUSE

I took the rocket home, and after analyzing the data and pictures of the wreckage I was able to determine the root cause of the problem: the separation point for the main parachute was in the wrong place. The first time around I had accounted for the length of the shock cord but not for the shroud lines attached to the parachute. I believed that the shock cord alone would be enough to pull the main chute out of the airframe, but the shroud lines got pulled first, and their length acted to disperse the pull on the parachute, so the canopy never cleared the upper airframe, which was still attached to the nosecone.

Nothing needed to be stronger. I restaged the recovery train so that the main parachute sits directly behind the nosecone, and at separation the nosecone pops off and frees the canopy directly.

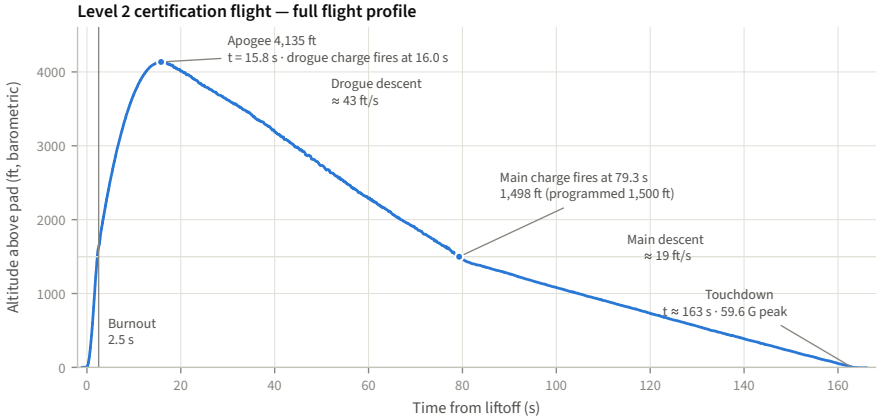


How the recovery train changed between the two Level 2 attempts. Schematic, not to scale.

L2 — SECOND ATTEMPT AND CERTIFICATION FLIGHT

The second attempt is the one that earned my Level 2: a J250 motor to 4,135 feet, main deployment at 1,498 feet, and a descent rate of 19 ft/s, right on target. As an additional testament to the design and construction, the rocket stayed within 6.8 degrees of vertical at burnout, meaning that it flew straight up until burnout. The recovery system did exactly what I designed it to do. The main parachute deployed at 1,498 feet, the altitude I had programmed it to, and brought the rocket down at 19 feet per second.

The electronics bay I designed and 3D-printed protected the altimeter and GPS through launch, both ejection charges, and a 59.6 G touchdown, and the GPS tracker maintained tracking through recovery.



Barometric altitude for the whole Level 2 certification flight, with the events the altimeter logged: burnout at 2.5 s, apogee at 15.8 s, main charge at 79.3 s and 1,498 ft, touchdown at about 163 s.

HOW CLOSE THE PREDICTION WAS

I had simulated the flight in OpenRocket before I flew it, and afterward I could check the prediction against the record. The model said 4,561 feet and 705 feet per second, and the rocket flew 4,135 feet and 658 feet per second, so I was 9 percent high on altitude and 7 percent high on speed. The landing velocity it predicted was 19.9 feet per second against the 19 I measured. I had not thought of it as a prediction-then-check exercise at the time, but that is what it was, and it is the closest I have come so far to knowing how a vehicle would behave before building it.

WHAT I LEARNED

The failure on the first attempt taught me to be more thorough in my designing but more specifically my testing. This kind of mistake could have been avoided through more rigorous testing and by verifying that the parachute gets pulled out during separation testing on the ground. All in all, what has stayed with me is this: failure is a lesson waiting to be discovered. I prepared what I could, but it was watching the failure of my rocket that taught me the final lesson.



The Level 2 certification flight leaving the pad, 8 August 2026. It reached 4,135 feet.



Found on the lakebed after the Level 2 certification flight, 8 August 2026, about six minutes after launch.



The same vehicle in hand after the Level 2 flight, airframe and plywood fins intact. The first attempt came back with three of its four fins damaged.

INDEPENDENT PROJECT · ROCKET 2 · AUGUST 2026

Supersonic flight test and failure reconstruction

Scratch-built Blue Tube airframe, 3.1 in OD / 3.0 in ID, 82 in long · 54 mm fiberglass motor mount · AeroTech L1000W, ~2,714 N·s · fiberglass ogive nosecone · fiberglass clipped-delta fins, 5.00 in root chord, 0.25 in thick · dual-deploy recovery · Featherweight Blue Raven 2.0 · Eggfinder GPS · two flights

ROLE	Sole designer, builder, test operator, and analyst. The airframe, fins, and electronics bay are my design and my build, the altimeter is off the shelf, and the tracker is a kit I soldered.
STATUS	Redesign specified · rebuild in progress · not yet re-flown
TOOLS	SolidWorks · OpenRocket · FDM 3D printing · Soldering (Eggfinder GPS / TX / RX kits) · Blue Raven 2.0 altimeter setup · Ejection charge sizing · Flight data reduction · Failure investigation

1,367 ft/s
peak vertical velocity · ≈ Mach 1.2

2.20 s
time from liftoff to breakup

3 → 88 G
side load growth in ≈ 0.35 s

535 → 290
roll rate decay, deg/s, before rupture

500 Hz
inertial record recovered intact

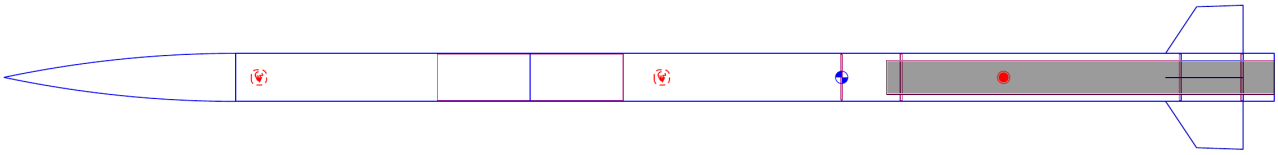
My second rocket reached Mach 1.2 and broke apart in flight. I recovered the electronics bay intact, pulled the full 500 Hz flight record off it, and spent weeks working out from the data and the wreckage that the fins failed first and the airframe opened afterward.

OBJECTIVE

Design and build a rocket from scratch capable of passing Mach 1, fly it supersonic, and recover it with a complete flight record on board. Prove the airframe, the recovery system, and the electronics on a subsonic flight first, before risking the vehicle on the supersonic attempt. Add two successful flights to the NAR Level 2 HPR Flight Log.

THE VEHICLE

I built this one straight after earning my Level 2 certification. I designed the vehicle from scratch around a 3.1 inch Blue Tube airframe and a 54 mm fiberglass motor mount, with a dual-deploy recovery system and a custom 3D-printed electronics bay holding the Blue Raven altimeter and the Eggfinder GPS tracker. I checked the design in OpenRocket, which put the static margin at 3.37 calibers and predicted 1,497 ft/s on an AeroTech L1000W, ~2,714 N·s, sized the ejection charges for the airframe volume, and sourced custom G10 fiberglass centering rings for the airframe.



The vehicle as I drew it in OpenRocket: fiberglass ogive nosecone, 19 in upper body tube, the coupler and avionics bay, a 48 in lower body tube over a 54 mm fiberglass motor mount, and four fins. The blue quartered marker is the center of gravity and the red one the center of pressure, 3.37 calibers apart.

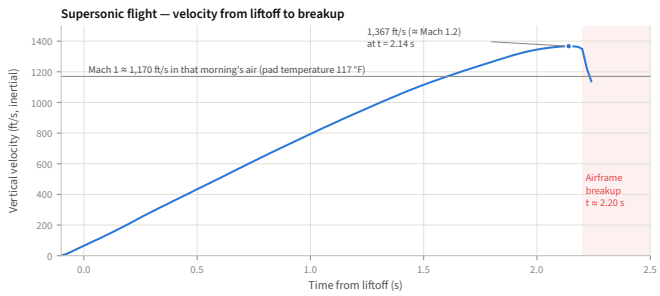
THE STAGED APPROACH

Rather than flying the rocket straight to its design limit, I flew it on a smaller motor first as a subsonic shakedown. That flight reached 2,833 feet, separated at apogee, and deployed the main parachute at 1,499 feet for an 18 ft/s descent, which confirmed that the airframe, the recovery system, and the electronics all worked before I risked the vehicle on a supersonic attempt.

Before either flight I ground-tested the ejection charges in the actual airframe, on 22 July, with the section resting on a stand and cushions set downrange to catch it. Four tests, all successful.

WHAT HAPPENED

One week later I flew the supersonic attempt. The rocket reached 1,367 feet per second, about Mach 1.2, at 2.14 seconds into the flight, and broke apart at about 2.20 seconds — still under thrust, 81 percent of the way through a 2.72 second burn, near maximum dynamic pressure. The avionics bay came down intact, and I was able to pull the complete flight record off it: 500 Hz inertial data and 50 Hz barometric data covering the whole flight.

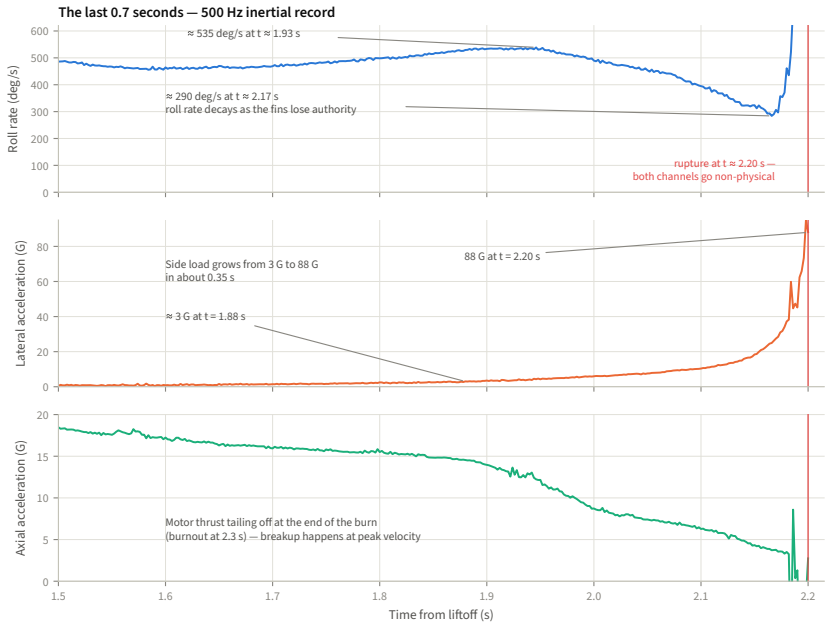


Vertical velocity from the inertial channel, liftoff to breakup. The rocket crossed Mach 1 at about 1.6 s and peaked at 1,367 ft/s at 2.14 s.

WHAT THE DATA SHOWED

I spent weeks going through that data and the wreckage. The data showed that side loading went from 3 G to 88 G in about a third of a second, the roll rate decayed from about 535 to 290 deg/s, and all four fins were missing from what I recovered.

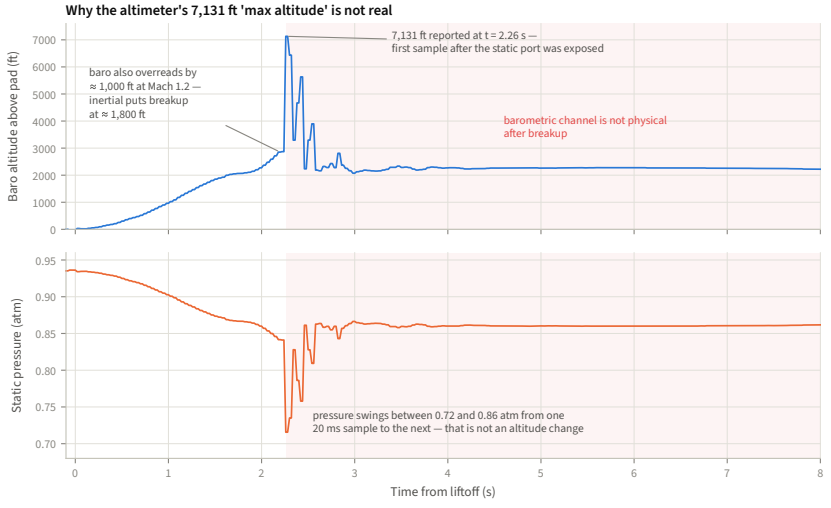
My first read was that the airframe wasn't strong enough — but the roll rate started dropping before the airframe came apart, which meant the fins failed first and the airframe opened afterward. I had to throw out my first explanation and go with what the data showed.



The last 0.7 seconds before rupture, from the 500 Hz record. Roll rate peaks at about 535 deg/s and then decays while the side load grows from 3 G to 88 G. At t ≈ 2.20 s both channels go non-physical, which is the airframe letting go rather than any sensor limit.

THE ALTITUDE THAT WASN'T REAL

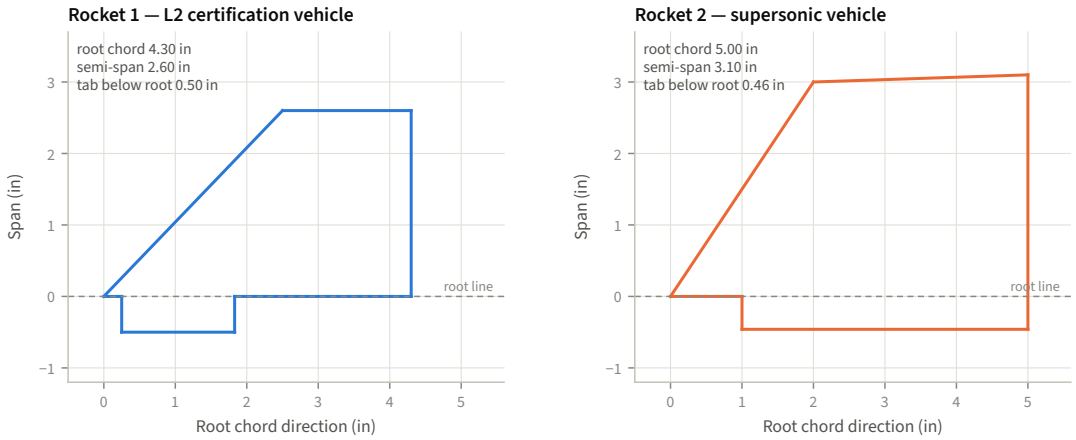
The altimeter's summary file reported a peak altitude of 7,131 feet, far above what actually happened. When the airframe opened, the static port was exposed, and the pressure reading started swinging between 0.72 and 0.86 atm from one 20 ms sample to the next, which is not an altitude change. By cross-checking it against the backup inertial data I was able to establish the real trajectory, and I threw the corrupted value out. The barometric channel was not trustworthy before the breakup either: at Mach 1.2 the pressure at a flush static port is not free-stream static, and that channel reads roughly a thousand feet high. Integrating the inertial record instead puts the breakup at about 1,800 feet.



Why the altimeter's 7,131 ft summary value is an artifact: the static pressure channel becomes non-physical the moment the airframe opens.

WHAT I CHANGED

I specified the redesign from what the data showed: thicker, stiffer fins with a stronger root attachment, and a composite airframe replacing the original tube. The changes I ended up making were materials and manufacturing decisions rather than changes to the shape of the rocket. The rebuild is in progress and has not flown yet.



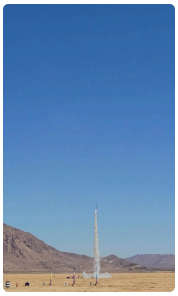
The two fin planforms. The supersonic vehicle's fin is a longer, taller clipped delta with a root tab that runs 4 inches along the airframe instead of 1.75.

WHAT I LEARNED

What has stayed with me from this flight is that the data has the final say. My first explanation was wrong, the number the altimeter handed me was wrong, and the only way to get to the real answer was to go through the record channel by channel and let it correct me. Typically, I design a part, build it, fly it, and use the data from that flight to decide what needs to change for the next version. This flight is the clearest example I have of that loop.



The finished vehicle on 14 July 2026, a month before it flew. 82 inches on a 3.1 inch Blue Tube airframe, with a fiberglass ogive nosecone, four fiberglass fins, and a 54 mm motor mount.



The subsonic shakedown leaving the pad, 8 August 2026 — the flight that had to work before I was willing to put the vehicle through Mach 1. It reached 2,833 ft and recovered cleanly.



The Eggfinder receiver showing the distance out to the rocket on the lakebed, 8 August 2026. Receiver, transmitter, and GPS module are all kits I soldered and assembled.

BRONCO LAUNCH · BRONCO AVIONICS · CAL POLY POMONA · JOINED MARCH 2026 · CO-LEAD SINCE MAY 2026

Bronco Avionics — Co-lead, Launch Operations

Two-stage rocket aimed at the Kármán line · Bronco Avionics, Bronco Launch · sub-team of four · avionics bay hardware, procedures, and pre-flight checkout

ROLE	Co-lead, Launch Operations, with Gregorio Marin, responsible for the vehicle's flight electronics.
STATUS	Active · MTD3 in relaunch preparation · EXO 0 in preliminary design
TOOLS	Blue Raven 2.0 altimeters · Featherweight GPS · Avionics bay integration · Harness and terminal work · Deployment ground testing · CircuitPython · Procedure writing · Team leadership

2
stages on the vehicle

4
people on the sub-team, me included

30 Aug 2026
two-stage pop test passed

I co-lead the Launch Operations sub-team for a two-stage rocket aimed at the Kármán line. I lead the avionics section of EXO Bronco. I have replaced and integrated the avionics bay hardware, written the procedures the sub-team runs on, and led a two-stage deployment test that fired correctly on both stages. The flight code on the telemetry boards is a teammate's, and I maintain it rather than having written it.

THE PROGRAM

Bronco Launch is Cal Poly Pomona's high power rocketry organization, and it was called UMBRA until this year. Bronco Avionics is the project I work under, and it runs in conjunction with Project EXO Bronco, whose goal is to fly a two-stage rocket to the Kármán line at 100 km. Launch Operations is the sub-team inside Bronco Avionics that I co-lead with Gregorio Marin. The vehicle we have in hand is MTD3, which flew once in February 2026 and is being prepared to fly again. The next vehicle, EXO 0, is in preliminary design.

Four of us, me included, are responsible for the Blue Raven altimeters that fire the deployment charges on both stages and light the sustainer, the Featherweight GPS, the Sprinklemini telemetry boards, the ground stations, and the launch-day command sequence. The avionics bay itself is ours too, which means the harness, the terminals, the switches, and the batteries. I joined the sub-team in March 2026 and became co-lead in May.

THE POP TEST

On 30 August 2026 we ran a black powder deployment test on MTD3, preparing it to fly again, and both stages fired correctly. I worked from the Pop Test Procedure I wrote with Ria Malik, integrated the sleds onto the rocket after Greg re-harnessed them, and connected to the Blue Ravens to command the test.

That test is the one piece of this vehicle's electronics I can tell you has been exercised on real hardware rather than on a bench, and the one change of mine it exercised was the screw terminals.

THE TERMINALS THAT HAD CORRODED

The booster's main deployment channel had lost continuity. I traced it to the barrier screw terminal strips in the bay. They were not the wrong part for the job. They were simply corroded with age, so I replaced them like for like with new 12-position strips. That is a replacement and not a redesign, and I would rather call it what it is. It held through both charges on 30 August.

Three other changes are in the new bay and none of them have been tested yet: new batteries, a voltage regulator, and a bay layout that changed to fit the batteries. The layout came out of my meetings with the structures team, and structures holds the CAD for it.

THE BOARD THAT DID NOT NEED ERASING

The sub-team's procedure for reprogramming a Sprinklemini opened by erasing the board. In September I put one on the bench and checked what state it was actually in before doing that. It was already running CircuitPython 9.1.3, it had no boot.py on it, and it was neither locked nor broken, so erasing it would have wiped a working configuration to fix nothing. The replacement procedure I wrote runs about twenty-five illustrated steps and supersedes the older versions.

Checking that board is also how I found the flight code sitting in exactly one place. It had been on the board since 7 May with no copy anywhere else, which meant losing the board meant losing the program's flight software. I backed it up on 15 September.

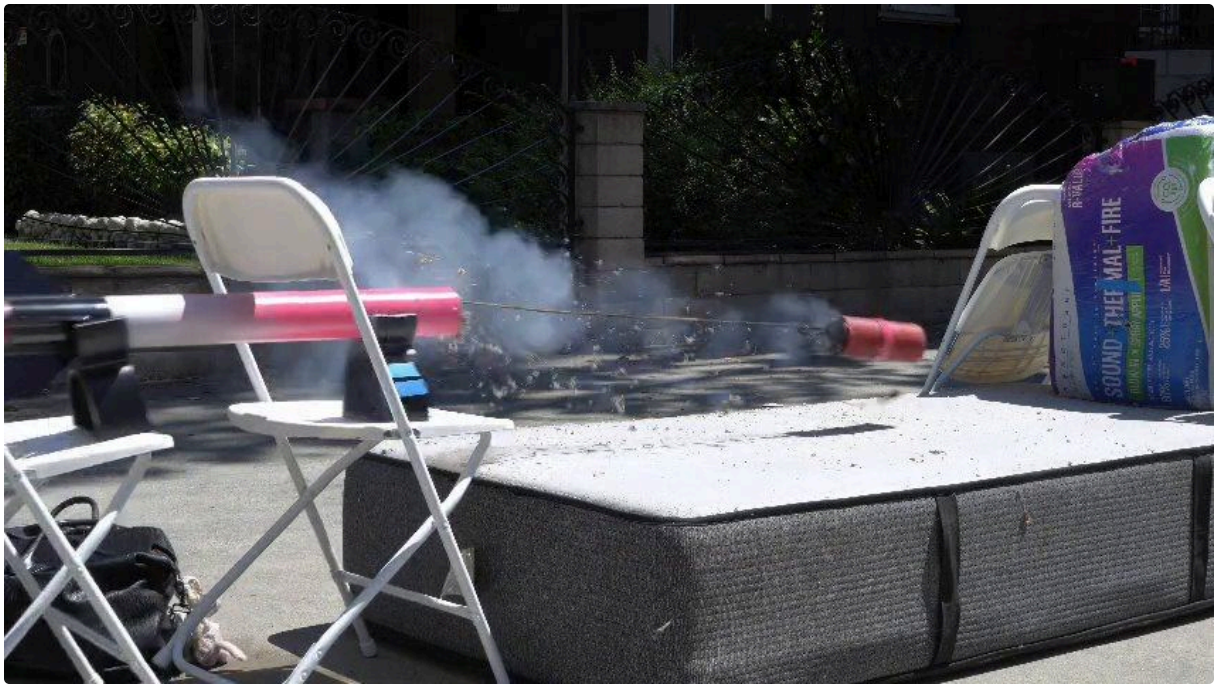
WHAT IS STILL OPEN

Two anomalies from the February flight kept MTD3 on the ground. The two Sprinklemini boards could not both be received at once, because both stages downlinked on 434.0 MHz. I documented that and reproduced it on the bench, where powering one board off made the other's data appear. The stages are now on separate frequencies, 433.0 for the booster and 434.0 for the sustainer, and in October both stages ran the full flight sequence at the same time on the bench, each to a separate ground station. The second is a Blue Raven that did not transmit its high rate record on one stage, and that is waiting on a re-test.

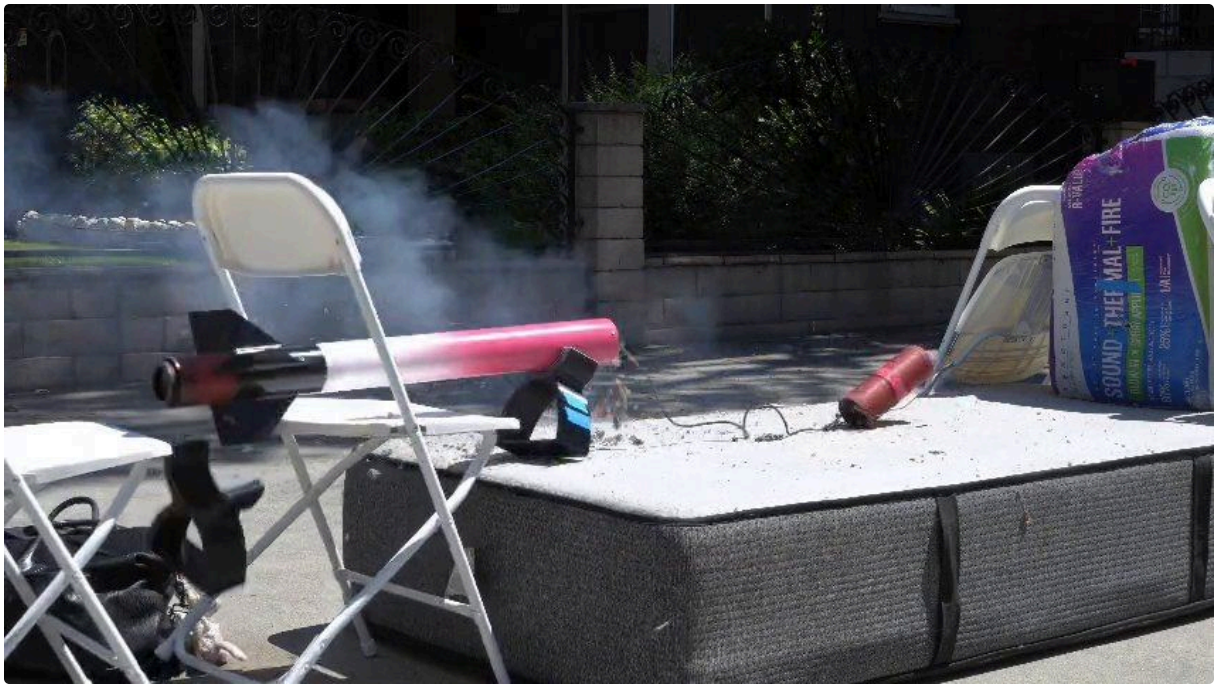
Neither of those is my failure. They were here before I was. They are what I inherited along with the hardware, and getting them closed is the job.

WHAT I HAVE WRITTEN

I set up the Drive structure the sub-team files into and the naming convention it runs on, and I wrote the avionics bay design constraints for EXO 0. The pop test procedure is mine and Ria Malik's, and the Sprinklemini reprogramming procedure is mine. The flight code on the Sprinkleminis is Tim's and Aivie's. I maintain it, document it, and work on it alongside Greg.



The booster ejection charge firing during the two-stage pop test on MTD3, 30 August 2026. The coupler is already out of the tube and the shock cord is taut. Frame from Bruce Duong's video of the test.



Three tenths of a second later. The booster section is coming down onto the mattress with the shock cord still attached. Frame from Bruce Duong's video.

Photographs and video of the pop test and of MTD3 are by Bruce Duong, the EXO Bronco project manager. Vehicle design details belong to the team and are shown here only with Bronco Launch's permission.

UMBRA, NOW BRONCO LAUNCH · CAL POLY POMONA · SEPTEMBER – DECEMBER 2025

Small Rocket Competition — sub-team lead

4-person team · 2,800 ft target altitude · flew 2,514 ft · Adafruit Metro Mini 328 and BMP390 logger on a micro SD breakout · 3D-printed nosecone, boat tail, and avionics bay · 29 mm AeroTech G64W

ROLE	Team lead. I ran the schedule and task assignment, reviewed and signed off the CAD before printing, harnessed the avionics bay, and wrote the logger code. A teammate, Eric, did the SolidWorks modeling — I did not know SolidWorks well enough at that point to do it myself.
STATUS	Flown · end of Fall 2025
TOOLS	Arduino C (Metro Mini 328, BMP390, SD logging) · Avionics bay layout and harnessing · Hardware fault isolation · OpenRocket · SolidWorks review and sign-off · FDM 3D printing · Team leadership and scheduling

2,514 ft
actual apogee against a 2,800 ft target

2nd
where we would have scored, before the disqualification

0.25 s
logging interval, time and altitude, to SD

4
people on the team

OBJECTIVE

Fly as close as possible to a 2,800 foot target altitude on the parts the competition supplied, and record altitude on board through the whole flight.

THE COMPETITION

The Small Rocket Competition was my first rocket. UMBRA gave each team a target altitude of 2,800 feet and a common parts list, and the team that came closest won. We were meant to be five people, but one dropped before the design phase, so we finished as four.

WHAT I DID, AND WHAT I DID NOT

I led the team. That meant running the schedule, deciding what had to be finished before each meeting, assigning the research and design work, and reviewing the SolidWorks files and giving the final go-ahead before anything went to the printer. Eric did the SolidWorks modeling. I did not know SolidWorks well enough at that point to do it myself, so my part was reviewing and approving it rather than drawing it. Once we reached construction, the four of us built the rocket together.

The two things that were mine end to end were the avionics bay and the code. I laid out the bay on a printed sled, harnessed it, held the boards down with rubber bands, and used quarters as ballast to balance the weight of the 9 V battery. The competition supplied the parts — an Adafruit Metro Mini 328 V2, a BMP390 pressure sensor, and a micro SD breakout — and I wrote the flight code that ran on them: about eighty lines of Arduino C that read barometric altitude and wrote a time and altitude pair to the SD card every 0.25 seconds, in a format that opened directly in Excel.

THE ALTIMETER THAT WOULD NOT WORK

My altimeter worked briefly on the day it was handed to us and then stopped. I took it home to fix it and could not get it to read at all. I worked through the wiring against the pinout, moved it from SPI to I2C, ran a bus scan that showed SDA and SCL both sitting high with no device answering, and stripped it back to the sensor alone with the SD breakout removed. The code compiled and ran, and the sensor was simply not there on the bus. I suspected the hardware rather than the code, but suspecting is not proving. A teammate suggested the test that settled it: run my code on another team's altimeter. It worked the first time. That isolated the fault to my hardware, UMBRA replaced the unit, and the replacement flew.

WHAT HAPPENED ON THE FLIGHT

The rocket flew on 23 November 2025 and reached 2,514 feet against the 2,800 foot target, measured by a separate altimeter the competition supplied. The logger I built recorded the whole ascent. On score we were first among the disqualified teams and would have placed second overall.

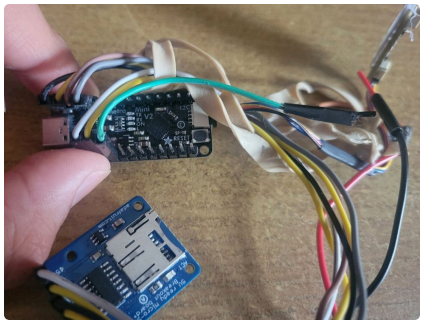
We were disqualified because the motor came out of the rocket. The boat tail was printed in PLA and doubled as the motor retainer, which meant a structural part with a job to do was sitting directly against the hot end of the motor casing. The heat softened it, the ejection charge then pushed the weakened plastic out of the way, and the motor slipped out of the back of the airframe. We found it later by walking the field, and nothing else on the vehicle was damaged.

WHAT IT LED TO

That failure is the reason every rocket I have built since uses a metal motor retainer threaded into the aft end, and the reason I treat any part that carries a load and sees motor heat as a structural problem before an aesthetic one. It is also the launch that got me into high power rocketry: by the end of the spring semester I had my Level 1 certification, and by August 2026 my Level 2.



The Small Rocket Competition vehicle at the launch site, 23 November 2025. Printed Haack nosecone, 2.2 in cardboard airframe, four birch plywood fins, and a printed conical boat tail at the aft end. Twenty-four inches from tip to tail.



The logger hardware, still on my desk: the Adafruit Metro Mini 328 V2 that ran my code and the micro SD breakout it wrote to, with the harness I made for them.

CAL POLY POMONA · FALL 2025

SR-30 turbojet performance test

Turbine Technologies SR-30 turbojet · test runs to 70,000 RPM · Excel data reduction · formal technical report

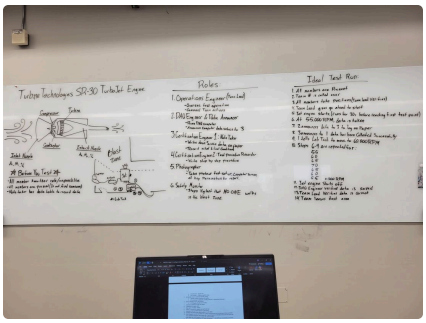
ROLE: Turbojet operator. The test plan splits the crew into six roles, and I ran the engine itself, then did the data reduction afterward. · STATUS: Complete · Fall 2025 · TOOLS: Test operation under written procedure · Instrumentation readings · Microsoft Excel data reduction and charting · Technical reporting

WHAT I DID

The SR-30 is a small turbojet engine on a laboratory test stand. The written test plan divides the crew into six roles, and mine was turbojet operator, which means I was the one running the engine. I ran it under the laboratory safety and startup procedures through test runs up to 70,000 RPM, then took the engine performance data into Excel and produced thrust and temperature trend charts for the formal technical report. That is the lab and test-stand side of my experience, running real hardware under a written procedure and turning the readings into a report.



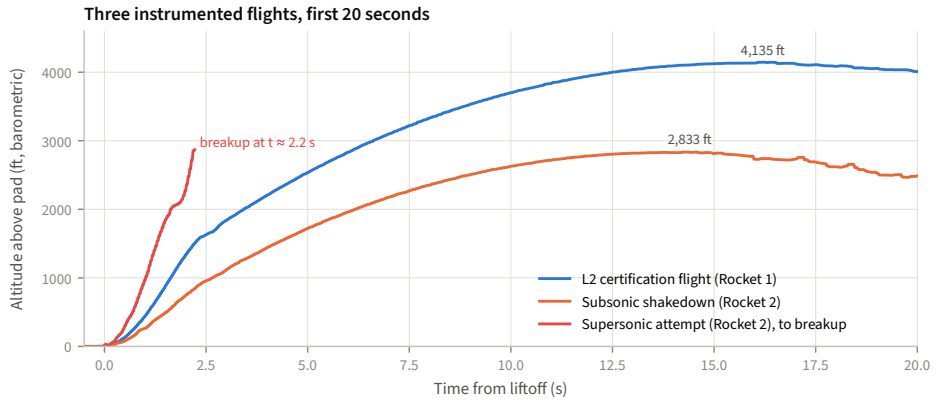
The SR-30 running in its test enclosure, November 2025. Hearing protection and the enclosure are part of the written startup procedure.



The test plan before the run: engine schematic, the six crew roles, and the fourteen-step ideal test run the team worked to.

Flight log

Every flight test I have flown, including the two that failed. The table is on the cover page.



The certification flight, the shakedown and the supersonic attempt on one axis. The supersonic attempt is shown up to the breakup, after which its barometric channel is not usable after that point.

Now

What I am working on this semester.

Rocket 2 rebuild

Thicker, stiffer fins with a stronger root attachment, and a composite airframe replacing the original tube. The changes are materials and manufacturing decisions rather than changes to the shape of the rocket. Not yet flown.

PIV design team — Design Liaison, since August 2026

I joined the design team for PIV (Particle Image Velocimetry) at Cal Poly Pomona as the design liaison, and I am responsible for creating and maintaining the team's Gantt chart. The main project is an airfoil with tubing at the leading edge that sucks or blows air to keep the flow attached, thinning the boundary layer and keeping it laminar. The goal is to finish the airfoil design by the end of the fall semester and begin testing in the spring.

ARO 2011L conceptual design project — Team Lead, Fall 2026

I lead a five-person team through a conceptual design of the Grumman F-14A Tomcat for Fundamentals of Systems Engineering & Design, presented as Grumman to NAVAIR in the 1968 VFX era. The deliverables build up into a Conceptual Design Review chart set and then a Preliminary Design Review.

Fall 2026 coursework

Engineering Statics · Materials Science & Engineering · Aerospace Computational Methods · Fundamentals of Systems Engineering & Design · Engineering Graphics.

Skills and tools

Each skill with the project I used it on. Finite element analysis is the next one I am learning.

CAD & SIMULATION

SolidWorks — part modeling, assemblies, drawings both rockets, electronics bays

OpenRocket — flight simulation and stability margin Small Rocket Competition, both rockets

AVIONICS & FLIGHT TEST

Dual-deploy recovery electronics and deployment logic Blue Raven 2.0 altimeter, Eggfinder GPS tracking

Ejection charge sizing and pre-flight checkout both rockets, EXO Bronco

Launch operations NAR Level 2 High Power Rocketry

PROGRAMMING

C++, Java, and MATLAB coursework

Embedded sensor and data-logging code Small Rocket Competition onboard logger

FABRICATION & HARDWARE

FDM 3D printing electronics bays for both high power rockets, nosecones, competition prototypes

Soldering and wiring harness assembly Eggfinder GPS / TX / RX kits, deployment charges, EXO Bronco avionics bay

DATA & ANALYSIS

Flight data reduction — 500 Hz inertial and 50 Hz barometric records supersonic failure reconstruction

Failure investigation from telemetry and wreckage two independent diagnose-and-redesign cycles

Microsoft Excel data analysis and charting · technical reporting SR-30 turbojet report

Requirements documentation · Gantt charts EXO Bronco, PIV

CERTIFICATIONS

NAR Level 2 High Power Rocketry Certification National Association of Rocketry · August 2026

NAR Level 1 High Power Rocketry Certification National Association of Rocketry · Spring 2026

About

I chose aerospace engineering because it's a place that lives on the frontier of human ingenuity. Here, one mistake can mean the difference between a flight that breaks into new territory and a failure. Speaking generally, I enjoy a good challenge. Coupling that with my interest in space and my logical nature, aerospace engineering was the natural path. Specifically, however, I like that “failure” isn't the end of an aerospace engineer's job, as failure can be reconstructed through analysis into a success — and that is something I have firsthand experience with.

Typically, I design a part, build it, fly it, and use the data from that flight to decide what needs to change for the next version. On my rockets I have been the one setting requirements and deciding when the project was good enough to proceed, and I have made those calls from test data, intuition, and analysis I could find online. Now I would like to be able to predict how a system will react before I build it, then test how the actual system works. Ultimately, I want to work on a project greater than myself — one I can help build up, learning how to be a proper engineer in the process.

Alongside school I have worked as a gurney and wheelchair transport technician for Transport Today since June 2024, moving non-emergency patients between hospitals, clinics, and care facilities and following the safety procedure through loading, transit, and unloading.

SCHOOL	California State Polytechnic University, Pomona
DEGREE	B.S. Aerospace Engineering · Expected May 2029
STANDING	GPA 3.97 · President's Honor List, 2025–2026
ORGANIZATIONS	Bronco Launch, Cal Poly Pomona's high power rocketry organization, formerly UMBRA · PIV (Particle Image Velocimetry) design team
COURSEWORK	Calculus I–III · Advanced Engineering Mathematics · Physics I (Mechanics) · General Chemistry
STATUS	U.S. citizen · eligible for a U.S. government security clearance
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